APPENDIX C

Strongly continuous semigroups

Throughout the whole section let $(E, \|\cdot\|)$ be a Banach space.

We will need some further basic functional analytic facts on linear operators A on the Banach space E. Mainly facts on the linear initial value problem

$$\begin{cases} \frac{d}{dt}u(t) & = Au(t)\,, t \geq 0 \\ u(0) & = u_0 \in D(A). \end{cases} \tag{C.1}$$

Here, $D(A)$ denotes the domain of A. If A is bounded (with full domain $D(A) = E$), a solution to (C.1) is given in terms of the operator exponential

$$u(t) := e^{tA}u_0 = \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k u_0, t \geq 0\,.$$

Since the most relevant operators are however unbounded, like operators involving the differential of a function, we need to find a suitable generalization. This is provided with the theory of semigroups.

Definition C.1. A family $(T_t)_{t\geq 0} \subseteq L(E, E)$ of bounded continuous linear operators on E is called a strongly-continuous semigroup (resp. C_0-semigroup) (of linear operators) if

(i) $T_0 = \mathrm{Id}_E =: I$,
(ii) $T_t \circ T_s = T_{t+s}$ for all $s, t \geq 0$,
(iii) $t \mapsto T_t u$ is continuous for all $u \in E$.

Given a C_0-semigroup we can define a linear operator A with domain $D(A)$ as the derivative of $T_t u$ in $t = 0$, if this derivative exists in E:

$$Au := \lim_{t \searrow 0} \frac{1}{t}(T_t u - u) \in E$$

with domain

$$D(A) := \left\{ u \in E : \exists \lim_{t \searrow 0} \frac{1}{t}(T_t u - u) \in E \right\}.$$

The linear operator $(A, D(A))$ is called the **infinitesimal generator** of the C_0-semigroup.

Remark C.2. (i) $(A, D(A))$ is densely defined, i.e., $D(A) \subseteq E$ is dense.
(ii) A is a closed linear operator, i.e., the graph

$$\Gamma(A) := \{(u, Au) \mid u \in D(A)\} \subset E \times E$$

is a closed subset in $E \times E$. This is equivalent to say that if a sequence $(u_n)_{n\geq 1} \subset D(A)$ converges to some element $u \in E$ and if in addition the

sequence $(Au_n)_{n\geq 1}$ is E-Cauchy, then this implies that $u \in D(A)$ und $Au = \lim_{n\to\infty} Au_n$ in E.

The resolvent set $\rho(A)$ of a closed linear operator A is defined as

$$\rho(A) := \{\lambda \in \mathbb{C} : \lambda I - A : D(A) \longrightarrow E \text{ bijective}\}.$$

For $\lambda \in \rho(A)$ the operator

$$R_\lambda := (\lambda I - A)^{-1} : E \longrightarrow D(A)$$

is well-defined and continuous (!) due to the closed graph theorem. R_λ is called the resolvent of A. The complement $\mathbb{C}\backslash\rho(A)$ is called the **spectrum** of A.

THEOREM C.3. *(Hille-Yosida theorem) Let $(A, D(A))$ be a closed (linear) Operator on E. Then the following statements are equivalent:*

(i) *A is the infinitesimal generator of a C_0-semigroup $(T_t)_{t\geq 0}$ with*

$$\|T_t\|_{L(E)} \leq Me^{\omega t}, \quad t \geq 0,$$

for given constants M, ω.

(ii) *$D(A) \subseteq E$ dense, $\rho(A) \supset]\omega, \infty[$ and*

$$\|R_\lambda^k\|_{L(E)} \leq \frac{M}{(\lambda - \omega)^k}, \quad k = 1, 2, ..., \lambda > \omega.$$

In this case

$$R_\lambda = \int_0^\infty e^{-\lambda t} T_t dt.$$

REMARK C.4. A C_0-semigroup $(T_t)_{t\geq 0}$ is called a C_0-semigroup of contractions if $\|T_t\|_{L(E)} \leq 1$ for all $t \geq 0$. In this particular case the Hille-Yosida theorem simplifies considerably, because in this case the following statments are equivalent:

(i) A is the infinitesimal generator of C_0-semigroup of contractions $(T_t)_{t\geq 0}$.

(ii) $]0, \infty[\subseteq \rho(A)$ and

$$\|\lambda R_\lambda\|_{L(E)} \leq 1, \quad \forall \lambda > 0.$$

EXAMPLE C.5. (i) **Laplace operator** on $E = L^2(R^d)$.

Let $W(t)$, $t \geq 0$, be the d-dimensional (continuous) Brownian motion. Then Itô's formula applied to some function $f \in C^2(\mathbb{R}^d)$ yields

$$f(W(t)) = f(W(0)) + \int_0^t \nabla f(W(s)) dW(s) + \frac{1}{2} \int_0^t \sum_{i=1}^d \underbrace{\partial^2_{x_i x_i} f(W(s))}_{=\Delta f(W(s))} ds.$$

If we define

$$T_t f(x) := \mathbb{E} f(W(t) + x) = \frac{1}{\sqrt{2\pi t}^d} \int_{\mathbb{R}^d} f(y) e^{-\frac{\|x-y\|^2}{2t}} dy$$

then $(T_t)_{t\geq 0}$ is a C_0-semigroup on $L^2(\mathbb{R}^d)$ and one can show that

$$Af := \frac{1}{2}\Delta f = \frac{1}{2} \sum_{i=1}^d \partial^2_{x_i x_i} f$$

is the infinitesimal generator, since

$$\begin{aligned}\frac{1}{h}(T_h(f(x)) - f(x)) &= \frac{1}{h}\mathbb{E}(f(W(h)+x) - f(x)) \\ &= \frac{1}{h}\mathbb{E}\Big(\int_0^h \nabla f(W(s)+x)dW(s) + \frac{1}{2}\int_0^h \Delta f(W(s)+x)ds\Big) \\ &= \frac{1}{h}\underbrace{\mathbb{E}\left(\int_0^h \nabla f(W(s)+x)dW(s)\right)}_{=0} + \frac{1}{2}\mathbb{E}\Big(\underbrace{\frac{1}{h}\int_0^h \Delta f(W(s)+x)ds}_{=\Delta f(W(h)+x)}\Big) \\ &\xrightarrow{h\searrow 0} \frac{1}{2}\mathbb{E}\left(\Delta f(W(0)+x)\right) = \frac{1}{2}\Delta f(x).\end{aligned}$$

The precise domain of the infinitesimal generator Δ is given as $D(\Delta) = H^{2,2}(\mathbb{R}^d)$, which is the (Hilbert-) space of all L^2-integrable functions having L^2-integrable weak derivatives up to second order.

For general $D \subset \mathbb{R}^d$, D nonempty and open, $m \in \mathbb{N}$ and $p \geq 1$, the space $W^{m,p}(D)$ denotes the Banach space of L^p-integrable functions f on D having L^p-integrable weak derivatives up to order m. $W^{m,p}(D)$ is called the Sobolev space of order m and integrability index p and it is a Banach space w.r.t. the norm

$$\|u\|_{W^{m,p}(D)} = \left(\sum_{|\alpha|\leq m} \|\partial_\alpha u\|^p_{L^p(D)}\right)^{1/p}.$$

For $p = 2$ $W^{m,2}(D)$ becomes a Hilbert space. It can be shown that $W^{m,p}(D) = H^{m,p}(D)$, where $H^{m,p}(D)$ is the closure of the set $\{u \in C^m(D) | \|u\|_{W^{m,p}(D)} < \infty\}$ w.r.t. norm $\|\cdot\|_{W^{m,p}(D)}$. We will use this alternative notation $H^{k,p}(D)$ throughout these lecture notes.

(ii) **Laplace operator** on $E = L^2(D)$, $D \subseteq \mathbb{R}^d$ open. Define the first exit time

$$\tau_D = \inf\{t \geq 0 : W(t) + x \notin D\}.$$

Then the family $(T_t)_{t\geq 0}$ of linear operators

$$T_t f(x) := \mathbb{E}(f(W(t)+x)|\tau_D > t), \quad t \geq 0$$

is a C_0-semigroup.

Also in this case $\frac{1}{2}\Delta$ is the infinitesimal generator with domain $D(\Delta) \subseteq H_0^{1,2}(D) \cap H^{2,2}(D)$, where $H_0^{1,2}(D)$ denotes the (Hilbert-) space of one times weakly differentiable functions f with f and all partial derivatives up to first order L^2-integrable and f vanishing on the boundary ∂D of D.

For general $m \in \mathbb{N}$ and $p \geq 1$, the space $H_0^{m,p}(D)$ denotes the closure of the set $C_c^m(D)$ of m-times continuously differentiable functions with compact support contained in D, w.r.t. the norm $\|\cdot\|_{H^{m,p}(D)}$-norm. It is called the Sobolev space of order m with integrability index p and vanishing boundary conditions. An alternative notation is $W_0^{m,p}(D)$.

Particular case $D =]0,1[$
In this case, the functions

$$e_k := \sqrt{2}\sin(\pi k x), \quad k = 1, 2, \ldots$$

are a CONS, consisting of eigenfunctions of Δ with corresponding eigenvalues $-\pi^2 k^2$. In particular,

$$T_t e_k(x) := e^{-\frac{t}{2}\pi^2 k^2} e_k(x), \quad k \in \mathbb{N}.$$

Hence, for general $f \in L^2(]0,1[)$ we then have the following **spectral representation** of the semigroup T_t

$$T_t f(x) = \sum_{k=1}^{\infty} \langle f, e_k \rangle_{L^2} e^{-\frac{t}{2}\pi^2 k^2} e_k(x).$$

In particular,

$$\frac{d}{dt} T_t(f)\Big|_{T=t} = -\pi^2 \sum_{k=1}^{\infty} k^2 \langle f, e_k \rangle_{L^2} e_k$$

and

$$D(\Delta) = \left\{ f \in L^2(]0,1[) : \sum_{k=1}^{\infty} k^4 \langle f, e_k \rangle^2 < \infty \right\}$$